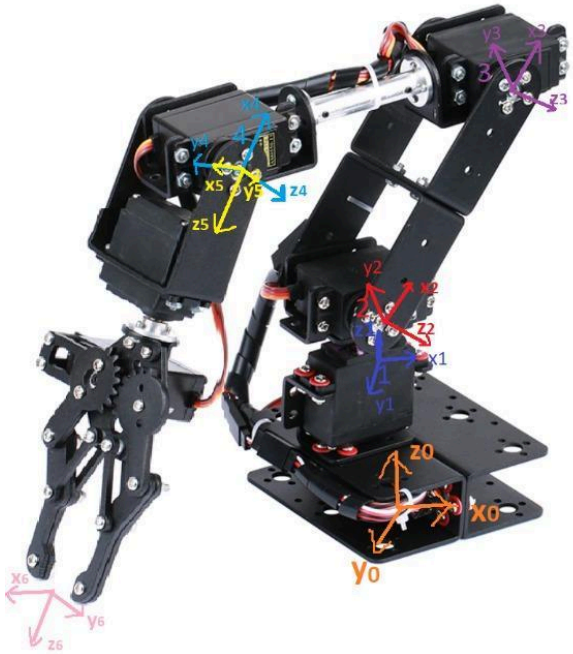
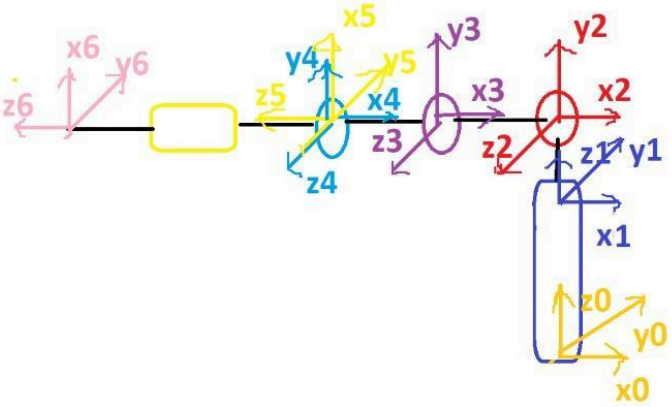
Colour Sorting Robot Arm

Alaa Elfiqi

Initial Specs :

This Project is aimed to function as a colour sorting robot arm. The robot arm is programmed to use; a camera for colour detection and image processing. Serial communication is then required to occur between the python code and the Arduino. The robot then uses forward and inverse kinematics in order to identify the angles for each joint for the gripper to be positioned in the right coordinates, and then the robot arm moves each coloured object from the pile of objects to the allocated section.





Functions Breakdown:

This project is broken into several main sections;

1. Camera colour Identification (via Iriun Webcam app)
    a. Blue, Green, Red and Yellow are the only registered colours. Any other colours will not be identified or sorted.
    b. Multiple coloured objects can be placed for the robot arm to pick and choose.
2. Python Programming - via Arduino program.
    a. Python code is required to use forward and inverse kinematics in order to identify the required angles of each joint to get the robot in the correct position for the gripper to be able to grip the coloured object.
3. Serial Communication then occurs between the code and the Arduino, sending through the needed information to correctly position the robot arm.
4. The Arduino uses the code and positions each joint at the required angle, enabling the robot to grab and sort the coloured objects.
5. Robot arm circuit to bring all of the functions together.

Implemented Sections:

Most of the required functions have been satisfied. The camera image processing is working exactly as needed. All of the four chosen colours are easily identified. The Python program successfully uses forward kinematics and calculates the required angles for each joint to be positioned to enable the robot arm to grab and move the coloured objects to the designated sections. The serial communication functions to optimum levels, being able to communicate to the Arduino all of the needed variables being the required joint angles, the colour allocation and the path of movement. All actions are successfully transmitted to the Arduino. All of these different functions combine and work together to create a system, enabling the robot arm to successfully complete the task at hand.

Non-Implemented:

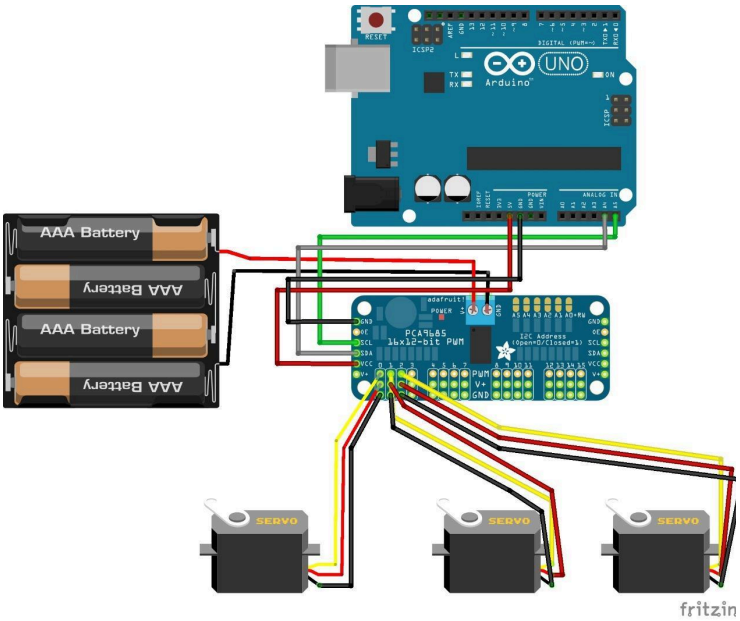
The python code was unable to do the inverse kinematic calculations in a proper manner and resulted in some faulty results. Hence the robot only used forward kinematic equations to identify the required angles. Another factor that the robot arm was unable to achieve is that the robot is not able to differentiate between different colours when placed next to each other. This means that only one coloured object can be placed for the robot to grab rather than having multiple objects and the robot picking and choosing.

Components:

The robot arm components are pretty simple, consisting of;

- 6 servos (Torque: 20 Kg.cm), (rotating: 180 degree)
- Power supply (5 V: 7 A)
- The electrical circuit of servos (16 Channel for servos)
- Camera
- Arduino board(Uno)
- Cubes(3.5*3.5*3.5*
- Boxes (10*10*15)
- Board to fix the arm on it (50*30)
- The length of the arm is 50.5 cm, the robot can get the cubic distance of 40cm

Circuit Wiring:



- The power supply of 5 volts is wired directly to the PCA 9685 which functions as a servo driver. Its role is distributing and allocating the multiple servo functions, which is not an available feature in Arduino UNO. Furthermore, the current provided by the Arduino is not enough to support multiple servo motors.
- The Arduino UNO is then connected to the PCA 9685 via 4 wires in order to transmit the commands and functions received from the python code in order to control the robot arm movements.
- The PCA 9685 then connects to the 6 servo motors that control the robot arm movement process.

Project Cost:

Below is a table outlining the cost of the project for each component.

Name of piece	Prices
Robot arm	1350
camera	500
Bord	150
Servo circuit	150
Power supply	300
Total	2500

Testing Procedure:

The testing procedure consisted of various different steps, each targeting a certain function of the robot. The first test was to identify whether or not the camera is able to identify each of the colours in a proper manner. The second part in the testing procedure is testing the code and the arduino. This section broke down into many different sub testing sections.

1. The first step was to calibrate the servo angles in order to ensure reliable outcomes.
2. Testing out simple movements in order to confirm the circuit wiring and connections are correct.
3. Testing out a single command without using the camera colour detection and manually inputting the colour required in order to test the movement and the joint angle calculations including the gripping feature.
4. Testing out the camera connection to the python code and robot colour identification feature.
5. Testing out the whole task process including colour detection, joint calculation, robot arm movement, gripping and colour allocating.

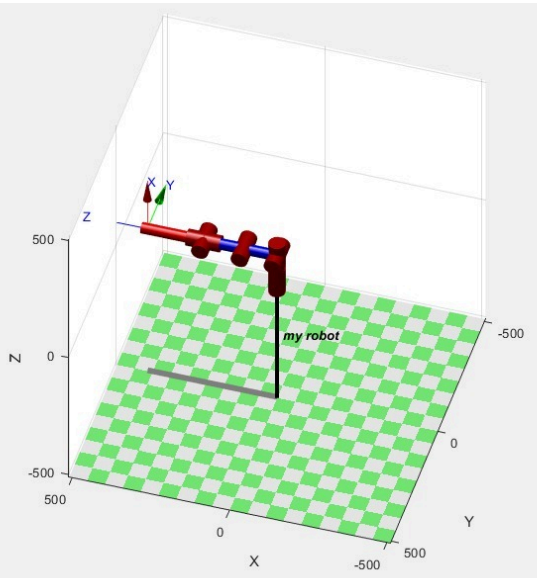
Testing Results:

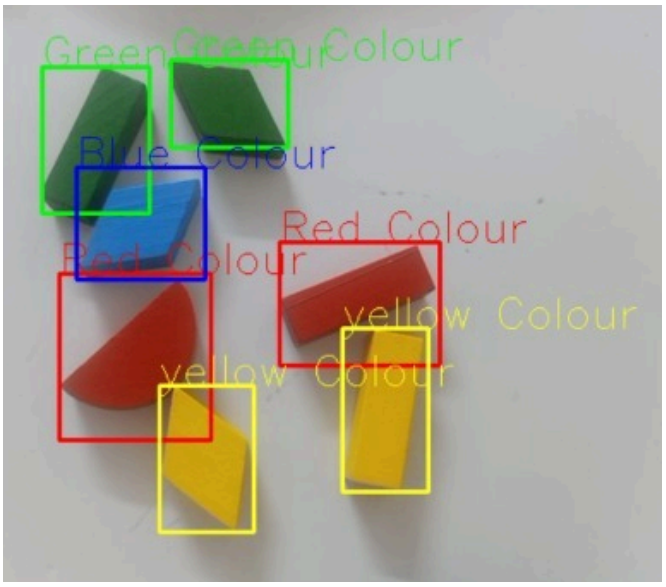
The testing procedure results were quite satisfying. During each step, various malfunctions and issues have occurred, however, they have been successfully troubleshot and fixed due to the organised manner in the testing procedure making the task of troubleshooting easy and organised.

Lessons Learned:

There are multiple important lessons that I have learnt during the process of this project completion;

1. Was able to form a connection between the theoretical work that I have learned and a practical aspect of it. I was able to see all of what I have learnt come to life in front of me.
2. I learnt how to calibrate the angles of the servo motors.
3. I was able to extend my knowledge regarding python coding such as
    a. Serial communications
    b. Using NumPy library
    c. Image processing
4. I was also able to extend my knowledge regarding Arduino based functions such as;
    a. Serial Communication
    b. Using Library of PCA 9685
5. Using Matlab Program to simulate the robot arm





Problems Faced:

The breadboard wires connecting the power supply and the PCA 9685 are not sturdy enough due to the high load which results in some slacking and non-functional actions. This malfunction affects various different aspects at random and tampers with all parts of the task making it difficult to identify and address the main issue.

Final Specs:

The final prototype of the robot arm meets most of the criteria set out at the very start. It is able to successfully identify the colour of a single object when placed in the scope of the camera as long as it meets the criteria of the four colours identified at the start. However, the robot arm is unable to successfully complete this process if there are multiple and differently coloured objects in the camera scope. This means that the robot arm at its current state can only identify the colour of a single object and move it once at a time. The python code is able to successfully conduct the forward kinematic calculations in order to identify the required joint angles and movement of the robot arm to complete the task at hand. Unfortunately, the python code still needs some editing in order to successfully complete the inverse kinematic equations in a correct manner.

Conclusion:

The Colour Sorting Robot Arm is considered to be a successful prototype. There are few future innovations that can be applied, however, it has served its main role and is extending my knowledge regarding many aspects.

Link to demo Video:

https://drive.google.com/file/d/1wjE8iimcxgBewZe0T8u2NEXRXe7dobWa/view?usp=sharing

Please find the different codes used in the document linked.